

1 HEADQUARTER AT DTBR

SITE CONDITIONS

Western Orientation:

Views to MS River and indoor light conditions in the afternoon (hotter time of the day).



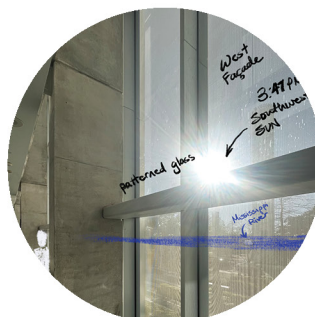
Floor 6 - roughly 86 ft



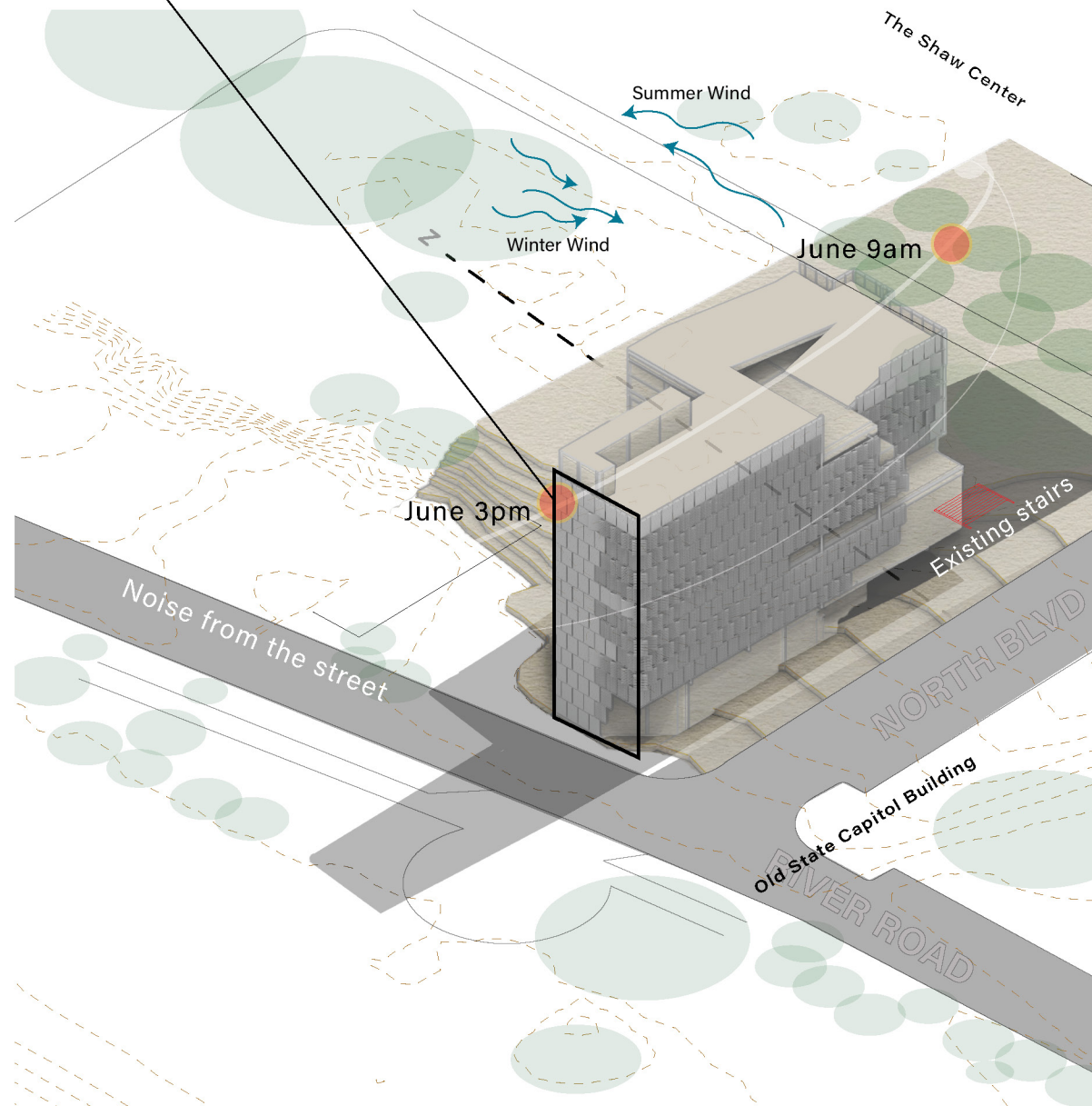
Floor 5 - roughly 74 ft



Floor 4 - roughly 62 ft

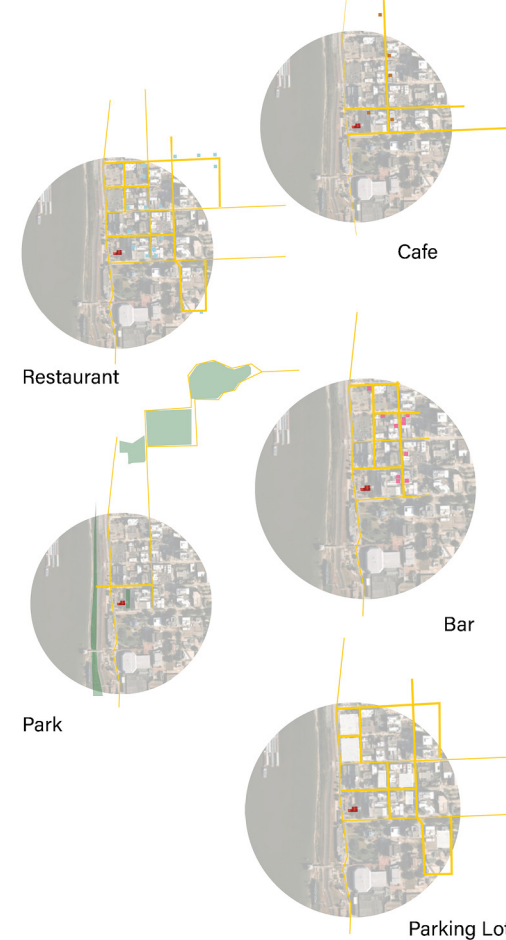


Floor 3 - roughly 50 ft



The project is located at downtown Baton Rouge, Louisiana, adjacent to the levee and the urban landscape. The project is commissioned by a national organization to design a Satellite headquarter in the South.

ACCESS and VICINITY



Spring (Mar.-May)

Temperature: 60°F (morning), 80°F (afternoon)
Wind: 5-10 mph, south/southeast
Average Precipitation: 4-5" per month
Humidity: 60-80%

Summer (Jun.-Aug.)

Temperature: > 75°F (morning), > 90°F (afternoon)
Wind: breeze, south/southeast
Average Precipitation: 6-7" per month
Humidity: >80%

Fall (Sept.-Nov.)

Temperature: > 60°F (morning), 80°F (afternoon)
Wind: 5-10 mph, north/northeast
Average Precipitation: 3-4" per month
Humidity: 50-70%

Winter (Dec.-Feb.)

Temperature: > 40°F (morning), 60°F (afternoon)
Wind: breeze, north/northwest
Average Precipitation: 3-4" per month
Humidity: 40-60%

2024 Spring Studio

Solo Project

Advisor: Robert Holton

Location: Baton Rouge, LA

Duration: 3 Months

Softwares: Rhinoceros, Grasshopper, CoveTool, Google Earth Pro, Adobe Suite (Photoshop, Illustrator, Indesign), Procreate



Gym
Natural light, open view,
egress access



Workspace
Natural light, open view,
egress, work-supporting
spaces direct access



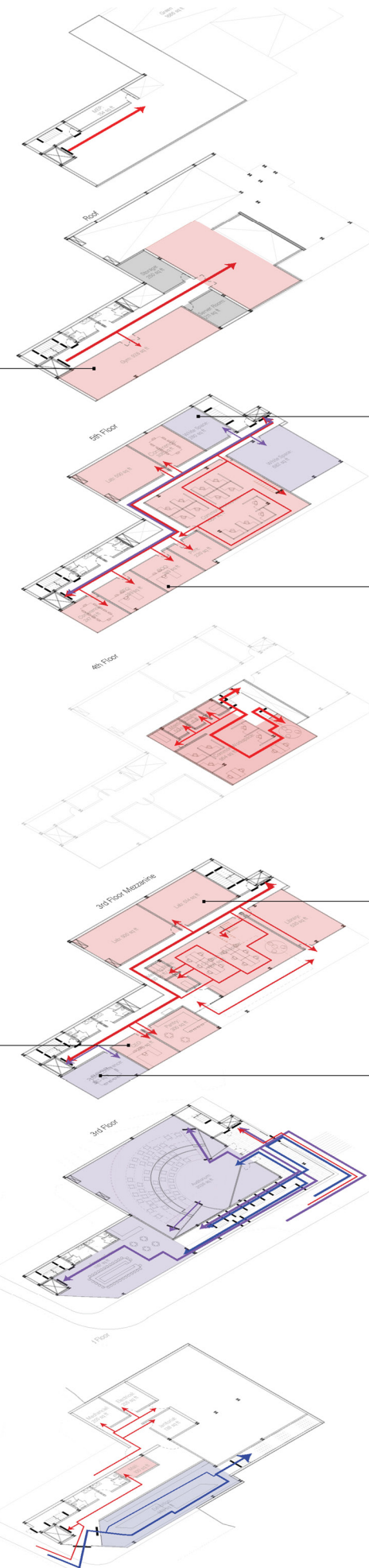
Admin. Office
Direct access to vertical
circulation, open view



Entertainment, Assembly
Natural light, open view,
egress, circulation between
exhibition, catering, and
auditorium



Entertainment, Exhibition,
Maintenance
Natural light, open view, egress,
circulation between exhibition,
catering, and auditorium



Flexible Space
Close to shared spaces and
vertical circulations



Admin. Office
Direct access to workspaces
(hallway and vertical
circulation), open view

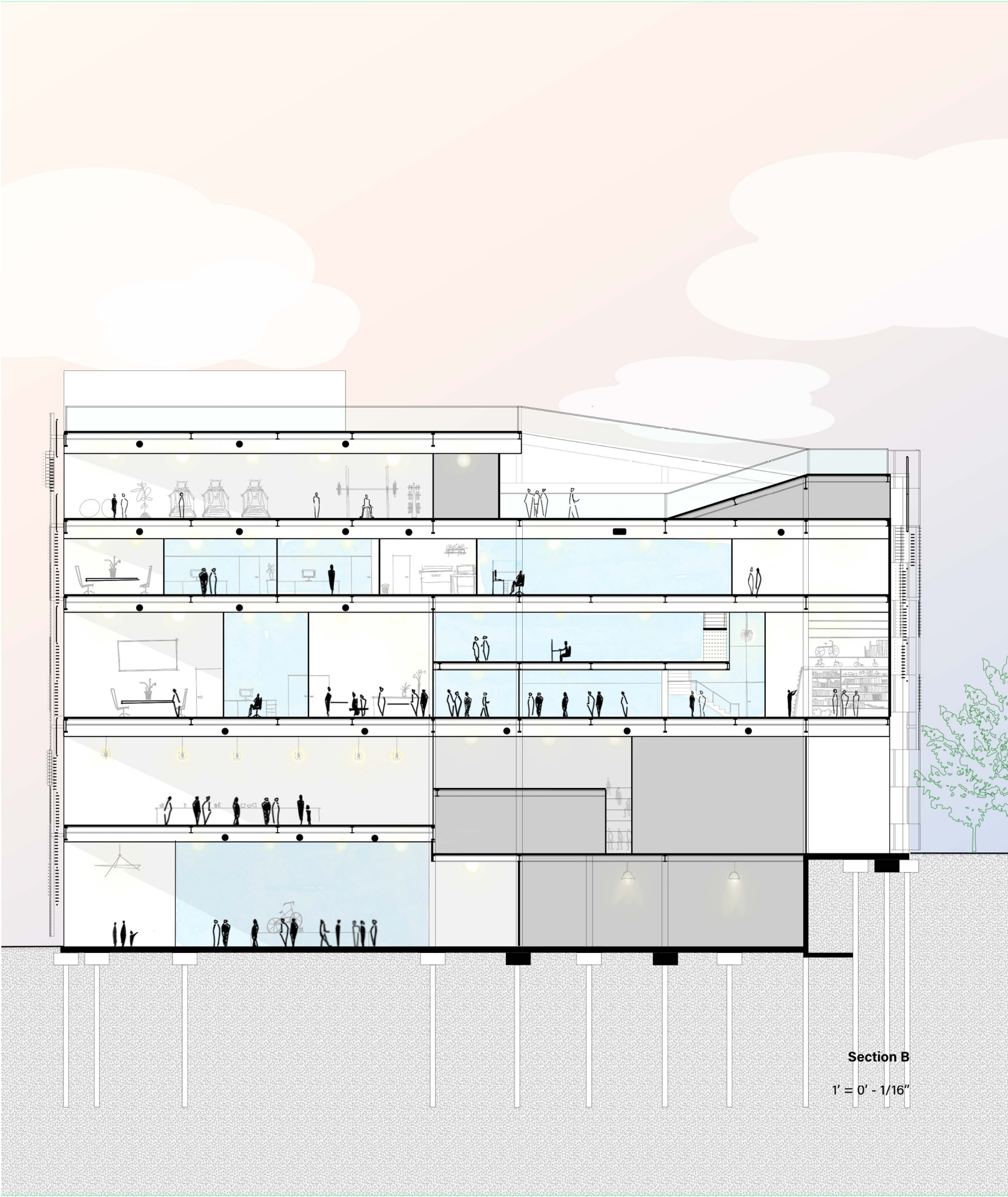


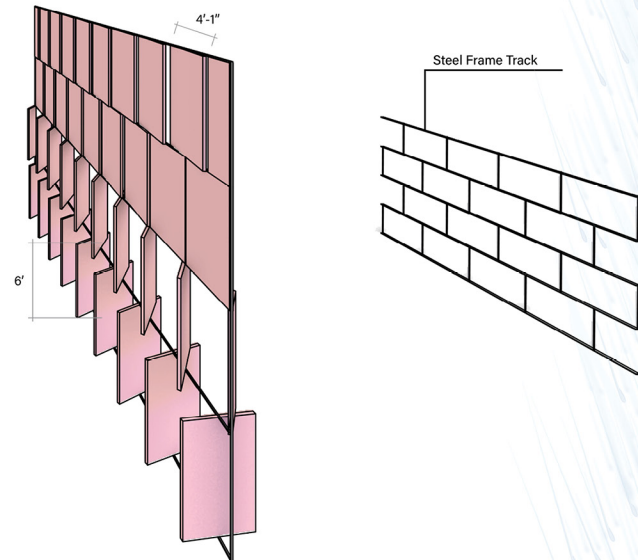
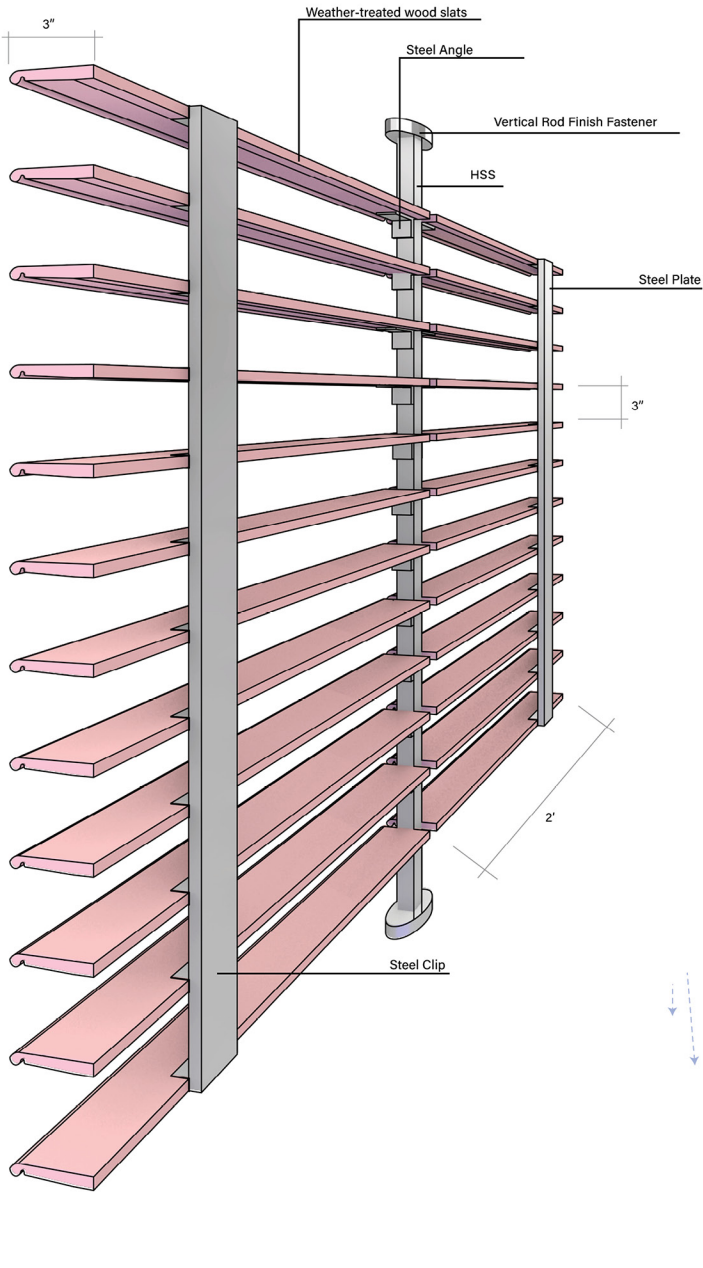
Close to workspace



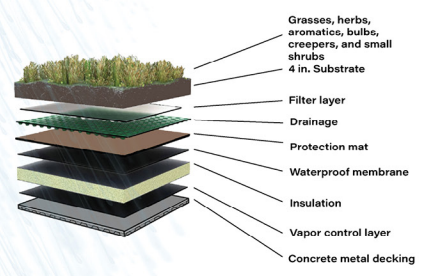
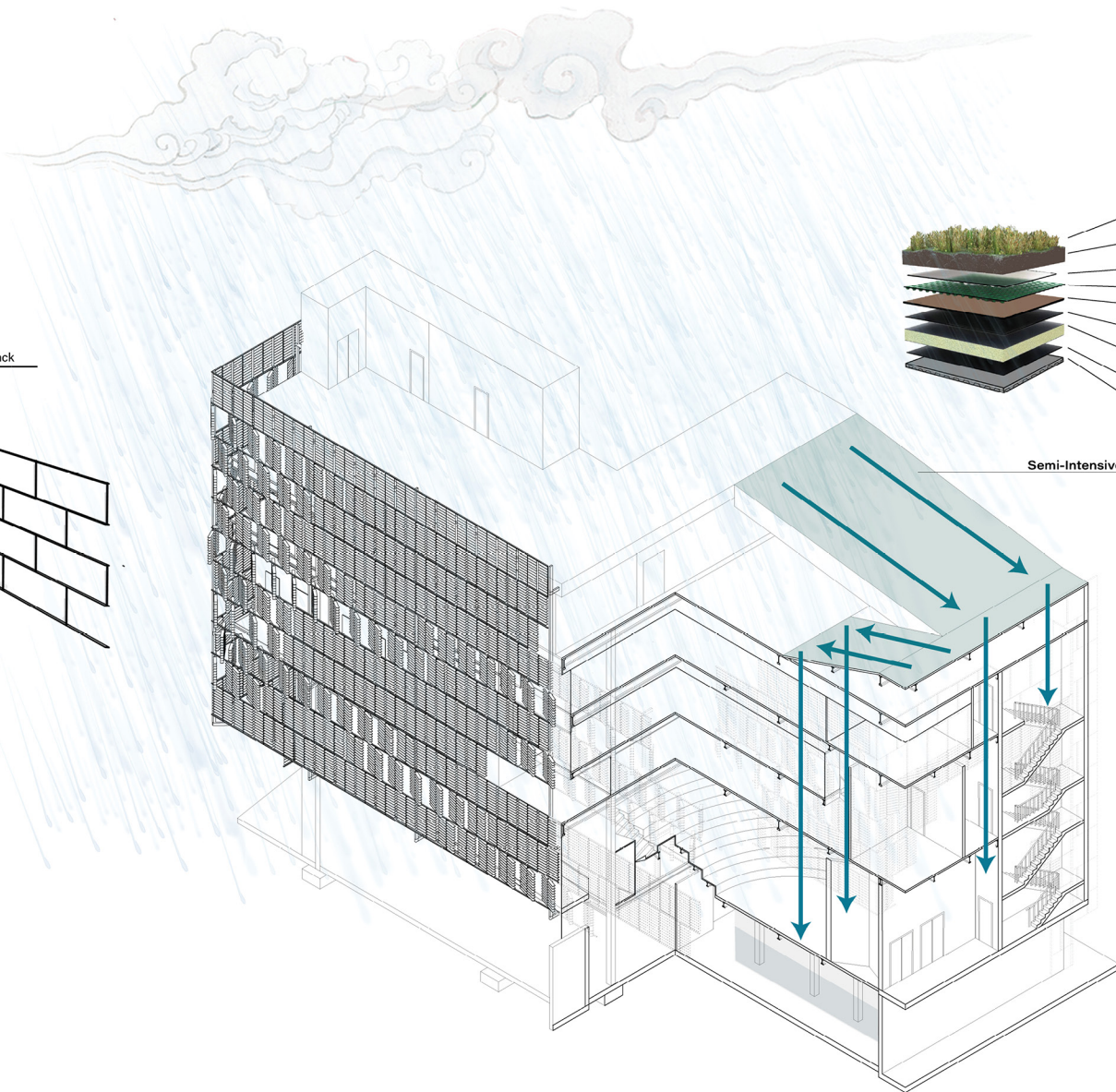
Conference
Close to vertical circulation
for employees and visitors

Visitors and public amenities are located on the bottom floors of the building. The amenities and activities on the upper floors are dedicated to Full-Time Employees.





Storm water Resiliency and Water Demand Diagram



Storm water Runoff
4.8 inches for 1 hours duration of a 100-year return period per IPC 2018 Appendix B, Figure 1106.1

Runoff Resiliency
- Infiltration and Evaporation through semi-intensive green roof
- Non-potable water recycling with cistern

Building Water Demand
1,755,000 gal./year = 260 business days x 6,750 gal./day
= 6,750 Gallons per day
750 Gallons for Business workers per day (15 gal/person x 50 people)
+ 1250 Gallons for Auditorium per day (5 gal/person x 250 people)
+ 1000 Gallons for Labs per day (100 gal/person x 10 people)
+ 3750 Gallons for Exhibition per day (75 gal/person x 50 people)

per Table 20.2 Planning Guide for Water Supply

Non-Potable Water Catchment System
Non-pervious roof
Total Gallons Precipitation Caught per year =
Catchment Roof Area x Gallons per square foot
8876.85 Gallons/year = 2863.5 sq.ft. x 3.10 Gallons/sq.ft
per Office of State Engineers TABLE 3-2: Approximate Annual Supply from Roof Catchment

Green Roof
Runoff = area x runoff coefficient for steep slope x rainfall depth
245 cf = 1686 sq.ft. x 0.35 x 0.42 ft
245 cf = 1832.73 gal.
Total retaining capacity = Total Gallons Precipitation/sq. ft./ - Runoff
3393.87 gal. = 5226.6 gal. - 1832.73 gal.

Total Non-Potable Water:
1832.73 gal. + 8876.85 gal. = **10,709.58 gallons/year**

